



The effect of AI-based CRM on organization performance and competitive advantage: An empirical analysis in the B2B context

Sheshadri Chatterjee^a, Nripendra P. Rana^{b,*}, Kuttimani Tamilmani^c, Anuj Sharma^d

^a Department of Computer Science & Engineering, Indian Institute of Technology Kharagpur, India

^b College of Business and Economics, Qatar University, Doha, P.O. Box-2713, Qatar

^c School of Management, University of Bradford, Richmond Road, Bradford BD7 1DP, UK

^d Chandragupt Institute of Management Patna (CIMP), Patna 800001, Bihar, India

ARTICLE INFO

Keywords:

AI-CRM

B2B

RBV

Institutional theory

Organization performance

Competitive advantage

ABSTRACT

Organizations have cultural-cognitive and regulative as well as normative elements that impact their employees. Organizations, by definition, cannot achieve a pure, stable state and always go through various change processes, both incremental and radical changes. Moving from legacy business-to-business (B2B) relationship management to an artificial intelligence-based customer relationship management (AI-CRM) is a gradual but paradigm change. AI-CRM leverages intelligent systems to automate the B2B relationship activities where the decision can be taken automatically without any human intervention. Relationship management in the B2B segment is considered a strategic activity of an organization. Moving from legacy to AI-CRM to facilitate B2B relationship management activities is an important decision, and proper implementation of AI-CRM is a critical success parameter for an organization. This study combines institutional theory and the resource-based view (RBV) in B2B relationship management to understand how AI-CRM could impact the firm's performance with varied firm size, firm age, and industry type.

1. Introduction

Artificial intelligence (AI) is on course to disrupt marketing management offering new prospects and challenges for marketing (Cao, Duan, Edwards, & Dwivedi, 2021; Davenport, Guha, Grewal, & Bressgott, 2019; Kumar, Dwivedi, & Anand, 2021; Mustak, Salminen, Plé, & Wirtz, 2021; Rust, 2020; Vlačić & Corbo, 2021). Especially, within the business-to-business (B2B) and industrial marketing segment, the applicability of AI-based solutions is well acknowledged and discussed (Bag, Gupta, Kumar, & Sivarajah, 2021; de Jong, de Ruyter, Keeling, Polyakova, & Ringberg, 2021; Martínez-López & Casillas, 2013; Paschen, Kietzmann, & Kietzmann, 2019; Paschen, Paschen, Pala, & Kietzmann, 2020). B2B marketers need intelligent solutions to automate the process of structuring, standardizing, aligning, and customizing data in a complex business environment (Farrokhi, Shirazi, Hajli, & Tajvdi, 2020; Fensel et al., 2001; Jabbar, Akhtar, & Dani, 2019; Syam & Sharma, 2018).

The impact of managing long-term customer relationships on business profitability has presumably influenced the academicians and practitioners to develop more understanding about customer

relationship management (CRM) (Richards & Jones, 2008; Wilson, Clark, & Smith, 2007). In particular, Richards and Jones (2008, p.121) provide a holistic definition of CRM as “a set of business activities supported by both technology and processes that is directed by strategy and is designed to improve business performance in an area of customer management”. In the B2B context, CRM helps organizations in identifying customers, understanding their requirements, developing customer knowledge, and building a profit maximization portfolio by establishing deeper buyer–seller relationships (Hart, Hogg, & Banerjee, 2004; Saura, Palos-Sanchez, & Blanco-González, 2019; Zablah, Belenger, & Johnston, 2004).

CRM integrates and analyses customer data generated from formal and informal interactions among the stakeholders, including the supplier and the customer. It also builds and maintain a profit-maximizing portfolio of customer relationships (Zablah et al., 2004). The integrated data from CRM is a living record of the firm's effectuated correspondence with its customer that is crucial to ascertain the customer's actual needs and lead management decisions (Ascarza et al., 2018; Kim, Park, Dubinsky, & Chaiy, 2012; Stein, Smith, & Lancioni, 2013). In B2B relationships, customers' data becomes vast and complex since it is

* Corresponding author.

E-mail addresses: nrananp@gmail.com (N.P. Rana), f09anujs@iimidr.ac.in (A. Sharma).

<https://doi.org/10.1016/j.indmarman.2021.07.013>

Received 30 January 2021; Received in revised form 26 July 2021; Accepted 28 July 2021

Available online 2 August 2021

0019-8501/© 2021 Elsevier Inc. All rights reserved.